



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc EXAMINATION
COLLEGE: SCIENCE AND TECHNOLOGY
SCHOOL: NATURAL AND APPLIED SCIENCES
DEPARTMENT: COMPUTER & INFORMATION SCIENCES
SESSION: 2019/2020 SEMESTER: ALPHA
COURSE CODE: CSC 214 CREDIT UNIT: 3
COURSE TITLE: HIGH PERFORMANCE AND DATABASE MANAGEMENT SYSTEM
INSTRUCTION: ANSWER ANY FOUR QUESTIONS
TIME: 2 1/2 HOURS

- 1(a) Explain (5) five costs or risks associated with database approach. [5 Marks]
1(b) Explain (3) differences between user views, a conceptual schema, and an internal schema as different perspectives of the same database [6Marks]
1(c) Transform the diagram below to a relational schema that shows referential integrity constraints. [6.5Marks]



- 2 (a) Explain the following: (i) Candidate key (ii) Foreign keys [2 Marks] (10)
2 (b) Use the following business rules to create the ERD for a small consulting company named ConsultCo. Show all **entities, relationships, optionalities and cardinalities**.
A client can contract many projects, but each project is contracted by one client. Some clients in the company client list have not (yet) contracted any project(s). A project may have many (consulting company) employees assigned to it, and each employee may be assigned to many projects. Some employees are not assigned to any project. One employee leads each project. An employee may lead more than one project at a time. One employee coordinates each project. An employee may coordinate more than one project at a time. Some employees never lead or coordinate projects. Many of the consulting company employees have multiple job skills that are applicable to the consulting work. Each job description- system analyst, database designer, accountant and so on- can fit many employees. Some job descriptions are not yet filled. Some employees do not have a specific job classification. [12.5 Marks]

2 (c) Define the following terms; (i) Data Model (ii) Database Instance (iii) DBMS

[3 Marks]

3 (a) Given the Patient_Appointment Table below

Candidate keys:

PNo, ADate (Primary key)

ADate, ATime, SurgeonNo

StaffNo, ADate, ATime

(i) Draw the Functional dependency diagram.

[3 Marks]

(ii) Normalizing the table below to Boyce-Codd Normal form and Show the final decomposed relations with the data.

[12 Marks]

Table 1: Patient_Appointment Table

StaffNo	DentistName	PNo	PName	Appointment Date	Time	SurgeonNo
S011	Tony Smith	P100	James White	12-Sep-08	10.00	S15
		P105	Jill Bell	12-Sep-08	12.00	S15
S024	Hellen Chris	P108	Tom Fred	12-Sep-08	10.00	S10
		P108	Tom Fred	14-Sep-98	14.00	S10
S032	Rose Kelvin	P105	Jill Bell	14-Sep-98	16.00	S15
		P110	John Stone	15-Sep-08	18.00	S13

Functional dependency

PNo, ADate, → StaffNo, ATime, DentistName, SurgeonNo (pk)

PNo → PName (pd)

StaffNo → DentistName Td

StaffNo, ADate → SurgeonNo

Date, ATime, SurgeonNo → PNo, PName, StaffNo, DentistName (ck)

StaffNo, ADate, ATime → PNo, PName (ck)

3 (b) Compare the Database approach and the File based approach to data management

[2.5 Marks]

4. (a) Given the following relations;

Customer_t(Customer_ID, Customer_Name, Customer_Address, City, State, Postal_Code)

Orderline_t(Order_ID, Product_ID, Order_Quantity)

Order_t(Order_ID, Order_Date, Customer_ID)

Product_t(Product_ID, Product_Description, Product_Finish, Standard_Price, Product_Line_ID)

Write the SQL statements to answer the following queries

- List products, finish, and unit price for all desks and all tables that cost more than 300 in the PRODUCT view
- Which products in the PRODUCT view have a standard price between 200 and 300.
- How many different items were ordered on order number 1004
- Which products have a standard price of less than 275.
- What is the address of the customer named Home Furnishings. Use an alias NAME for customer.
- List the unit price, product name and product ID for all products in the product table.

- vii. What is the average standard price for product in inventory [14 Marks]
- 4 (b) Explain Database Manager and its Responsibilities [3.5 Marks]
- 5 (a) Define five types of semantic object. [5 Marks]
- 5 (b) List 10 components of a DBMS and Explain 5 of them [10 Marks]
- 5 (c) Draw the ER diagram to represent the following business rules.
A vendor builds multiple types of tablet computers. Each has a type identification number and a name. The key specifications for each type include amount of storage space and display type. The company uses multiple processor types, exactly one of which is used for specific tablet computer type; obviously, the same processor can be used in multiple types of tablets. Each processor has a manufacturer and a manufacturer's unique code that identifies it. [2.5 Marks]
- 6 (a) Differentiate between Information Efficiency and Information Effectiveness [3 Marks]
- 6 (b) Explain the 6 stages of Information life cycle [6 Marks]
- 6 (c) Briefly Explain the Database Life cycle [4.5 Marks]
- 6 (d) At a weekend retreat, the entity type PERSON has three subtypes: CAMPER, BIKER, and RUNNER. Draw a separate EER diagram segment for each of the following situations: (6marks)
- (i) At a given time, a person must be exactly one of these subtypes.
 - (ii) A person may or may not be one of these subtypes. However, a person who is one of these subtypes cannot at the same time be one of the other subtypes.
 - (iii) A person may or may not be one of these subtypes. On the other hand, a person may be any two (or even three) of these subtypes at the same time.
- At a given time, a person must be at least one of these subtype [4 Marks]